

Josh Important Questions-Class Discussion

Given a binary tree, how can we find the maximum element at each level and return a vector containing these elements?

Ans. Find the maximum element at each level of a binary tree and return them as a vector.

- Use a level-order traversal (BFS) to explore the tree level by level.
- Maintain a queue to keep track of nodes at the current level.
- For each level, track the maximum value encountered.
- Example: For the tree [1, 3, 2, 5, 3, null, 9], the max values are [1, 3, 9].
- Return the results as a vector or list.

Convert the Binary Search Tree (BST) to a Greater Tree.

Ans. Convert a BST to a Greater Tree by replacing each node's value with the sum of all greater values.

- Traverse the tree in reverse in-order (right, root, left) to accumulate values.
- Start with a variable to keep track of the cumulative sum.
- For each node, update its value to the cumulative sum and add its original value to the sum.

Q. Given a Binary Search Tree (BST), find if there exists a pair of nodes whose values sum up to a given target. The time complexity required for this solution is $O(n)$, and the space complexity should be less than $O(n)$.

Ans.

- Use an in-order traversal to get sorted values from the BST.
- Utilize two pointers: one at the start and one at the end of the sorted array.
- If the sum of the values at the two pointers equals the target, return true.
- If the sum is less than the target, move the left pointer right; if greater, move the right pointer left.
- Example: For BST with values [2, 4, 5, 7, 10] and target 9, return true (4 + 5).

Q. Leet code 1448

Q. Rotate the sub-list of a linked list from position M to N to the right by K places (**GFG**)

Q. Maximum Sum of Non-Adjacent Nodes (**GFG**)